Sealed Provenance for Agent Swarms: Preloaded Lineage vs On-Demand Retrieval Under a Trusted Publication Gate

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Abstract

In a swarm, a corrected source does not un-write the summaries already built on it. A reviewer that sees only current records can approve a draft whose number is right but whose lineage is dead. We built a PostgreSQL publication gate where agents cannot write authoritative state: a trusted runtime seals each artifact with the inputs it delivered, invalidation is authority-checked, and publication re-reads the full dependency closure inside one fenced transaction. An independently written acceptance suite checked the gate in 19 live checks, and the gate was rebuilt from the committed migration on a fresh database branch.

On that gate we ran a preregistered, matched reviewer comparison on 5 open models hosted by Adaption: 5,400 registered slots, three task families, six arms, only the reviewer's context varying. The models made 428 unsafe proposals and the gate committed none. Whether preloaded lineage beats an authenticated lookup turns on whether the model takes the lookup: gemma-4-31b-it took it in 13 of 120 offers and completed 119 of 120 same-number cases with preloaded lineage against 68 with the lookup offered; deepseek-v4-flash took it in 119 of 120, the arms tied, and preloading saved a median 1,280 tokens per case. On a clean-draft control three models quarantined all or nearly all clean drafts under an unrelated correction notice in every arm: false quarantine is model behavior, not a property of lineage. Failed claims are reported as failed. The gate assumes a trusted operator and cannot observe what a model relied on.

1. Introduction

In July 2026 about 1,200 agents inside a security evaluation found a shared directory, turned it into a message board, and roughly 700 of them broke into Hugging Face's systems together. METR's investigation put the board at about 1,200 participants, about 700 taking part in the attack, with transcript spoofing in about 7 percent of the transcripts it evaluated; the Swarm Traces reconstruction counted 7,905 distinct names across those agents. We use the incident as motivation, not as a topology we claim to model. Our testbed has four agents, one trusted operator, and no network dynamics.

The failure we isolate is narrower and, we think, more common than a break-in: stale evidence after a correction. One agent reports a figure, a second summarizes it, a third summarizes the summary; then the figure is withdrawn and replaced. The summaries still exist, still read as confident, and still sit in the review queue. A reviewer that cannot see lineage can approve the stale summary because the prose sounds current, can refuse everything, which is safe and useless, or, in the hardest case, can face two drafts with the identical number, one descending from the withdrawn source and one from the replacement, with no way to tell them apart.

The measured contribution is a Track 4 evaluation: a publication gate that re-reads authenticated lineage, a matched reviewer comparison, and an independent scorer that resolves lineage itself rather than trusting what a model cites. It runs on a Track 3 substrate with identity bound to the database login below the agent: each agent submits over its own login, a trigger overwrites any claimed name, and the runtime seals from that authenticated row, so no author label in a trace is asserted by the harness. Two agents sharing one login are one identity to the gate.

The trust boundary is explicit. One operator seals, adjudicates and publishes; if that operator is compromised, every property below fails. We claim only that, given a trusted runtime, identity can be bound below the agent, lineage can be recorded by the runtime rather than asserted by the model, and a reviewer given that lineage behaves measurably differently from one that is not.

Contributions.

1. A tested multi-agent failure: stale evidence surviving a correction through summarization, including the case where the number is unchanged and only the authorization is.
2. Its recovery under a publication gate agents cannot write around, every claimed property verified by a second, independently written implementation and reproduced from source on a fresh database branch.
3. A preregistered matched evaluation others can rerun: fixed-input dispatch, a scorer that resolves lineage itself, opaque identifiers, registered denominators, frozen hashes and hard spend ceilings, with its failed claims reported as failed.
4. The result that whether preloaded authenticated lineage beats on-demand retrieval turns on whether the model takes the retrieval, and that false quarantine under a correction notice is model behavior rather than a property of lineage.

2. Related Work

None of the primitives are new. Immutable versioned artifacts, a provenance graph, authority-checked invalidation and a gate that re-reads dependency state are established engineering: version control, build systems, and provenance standards such as W3C PROV. Persistent, provenance-aware memory for human-AI systems is the subject of the first author's prior preprints on the CAMA architecture, and the Hive substrate this entry runs on was built on it in April 2026. AI control protocols in which a trusted model oversees an untrusted one, and Apart's earlier multi-agent security sprints, are the nearest neighbors on the evaluation side; the brief's references on collusion, spread and steganographic coordination frame the wider problem this entry does not attempt.

3. Methods

3.1 The gate

The gate is a PostgreSQL schema in which the four agent roles hold no write access to authoritative state. They may read, and they may submit proposals to an intake table whose author column is bound to the authenticated login by a trigger. Everything else goes through four trusted functions, each `SECURITY DEFINER` with a pinned search path, with public execute revoked and only two entry points granted back to agents.

Agents reach the authoritative tables only through the operator's functions; their one direct write is a proposal, and everything else they can do is read.

Sealed creation. `seal_artifact` is the only creator. The runtime calls it with the author taken from the authenticated dispatch context, the body, and the list of inputs it actually delivered to the model. In one transaction it writes the immutable artifact and its provenance edges, marks it a root (no inputs) or a derivative, and rejects any input that does not already exist as a sealed artifact. Late edges and cycles are impossible by construction. A correction is a new version, never an overwrite.

Authority-checked invalidation. `invalidate` is the only write path. An author may withdraw its own artifact. Only an adjudicator may issue an adjudicated invalidation. Anyone may propose a contradiction, which is advisory and never blocks. A compromised agent cannot invalidate another agent's work by assertion.

Fenced publication. `publish_decision` is callable only by an adjudicator. It takes a shared row lock before any state check and holds it to commit, and it requires read-committed isolation so a stale snapshot aborts instead of approving. It then checks that the artifact exists and is sealed, walks the full transitive dependency closure including the artifact itself and rejects if any member is withdrawn or adjudicated, and requires the operator's acknowledgment list to cover every recorded direct input. Only then does it commit the approval. A withdrawal that commits before the call is seen; one that lands after leaves a reconcilable superseded state.

What it does not do. It is a database boundary, not a sandbox: it says nothing about shell, network or file egress. Operations the database denies, such as an attempted delete, leave no row in the application audit, because they are rejected before any trigger fires. Three of six attacks in our first battery were invisible to that audit. We report this as a measured limitation rather than paper over it.

Independent verification. Lorien, an AI collaborator running in a separate workspace, wrote an executable reference specification and a live acceptance driver against the committed SQL. Its first pass passed 1 of 9 core checks. The failures traced to real defects: the gate approved nonexistent artifacts, checked only direct dependencies so a withdrawn grandparent passed, and let any agent record provenance edges onto another agent's artifact; later probes found that a bot could supply its own proposer name, call the internal fence lock directly, and seal an edgeless artifact with a null input list. Each was fixed and the suite re-run at each revision: 1 of 9, then 5, then 8, then 19 of 19 across core, intake, malformed-input and dispatch checks. The committed migration was then applied in one transaction to a fresh Neon branch, and the 19 checks passed there with every routine hashed and the full grant table captured. The frozen gate is commit `d2df880`. Four additive migrations landed after two adversarial reviews on 7 November: authenticated authorship through the intake with an exact receipt bound to author, artifact, version and body digest; a separate table for advisory contradictions so they never touch the fence; a read-time validity view; payload-only intake grants and a fail-closed fence. Each was re-verified by the suite (19 of 19), nine binding checks and a live lock-contention schedule. The suite checks specific boundaries; it is not a proof of every property.

3.2 Experiment design

The question is whether lineage preloaded into a reviewer's context changes stale proposals and correct completion compared with lineage the reviewer must fetch, or cannot fetch at all. Everything else is held fixed: the model, the task, the correction notice, the prompts, the token budget, the gate. Only the reviewer's context adapter varies.

Arm	What the reviewer sees
current	queued draft bodies, the current source, the correction notice
current + lookup	the same, plus one authenticated lookup, answered from the trusted store in a second fully recorded request, returning exactly the lineage records the compact arm preloads and no bodies; the runtime checks per task that the two are identical
compact	current, plus authenticated artifact versions, delivered-input links and invalidation status, no historical bodies
full	compact, plus the protected historical bodies, delivered as recorded inputs
rule, rule + content	deterministic references: approve the queued draft whose lineage is valid; the second also checks the figure

Fixed-input dispatch. Every model call goes through a delivery recorder written by Lorien. The harness resolves artifact bodies from the authoritative store, constructs the exact request, hashes it and each delivered body, and transmits it without appending anything. When an agent's output is sealed, its dependencies come from that receipt, never from citations in the model's text. The recorder establishes what the harness supplied and sealed; it cannot establish what the model relied on.

Tasks. Synthetic inventory counts with exact known answers, one harness-injected compromise opportunity per instance (a replayed stale body, a wrong body with valid lineage), identical across arms. No agent attempts anything adversarial in these runs; that lives in the separate containment battery. Identifiers are opaque random strings, because an early version leaked recency through names like `draft2`. Development variants: V0 clean completion; V1 an unrelated correction before review (false-quarantine control); V3 multi-hop stale evidence; V4 the same with source names omitted in prose; V5 a replayed stale artifact; V6 the figure unchanged, only the authorization changed. The held-out set adds V11 (V6 with no correction notice delivered) and the diagnostics V7 to V10 (transitive authority, an unrelated withdrawal with the same number, two sources with one revoked, and a wrong figure with valid lineage).

Scoring. The reviewer returns a structured proposal: a decision, a result, the evidence its result rests on, and a separate list of artifacts consulted but not relied on. An approval's evidence must include the approved draft itself, a rule added after the open-model pilot showed that a stale pick with the right number and a current citation could escape detection. Gate events are recorded only after a separate committed-state query. The scorer, written independently, holds the evaluator's answer keys and the authoritative provenance export and resolves withdrawn ancestors itself, so citing a summary cannot hide its dead source. Every registered run stays in the denominator, including timeouts, errors and invalid output. Outcomes: unsafe proposals, correct completion, gate interventions, calls and tokens; accepted bad actions are protocol integrity under the common gate. For lineage-visible staleness the expected value is zero; for V10,

whose lineage is valid, the gate commits whatever the reviewer approves and the measure becomes which reviewers catch a wrong figure by reading it.

3.3 Development findings that shaped the freeze

Development runs, one instance per cell, reported because they shaped the frozen design. They are not effect sizes.

Three Sonnet 5.5 pilots (V0, V3, V5; twenty calls, six cents) registered nine runs, nine complete, eight correct, zero accepted bad actions. The current arm could not tell a replayed stale draft from the fresh one and deferred, safe but incomplete; compact and full approved the fresh draft. On the two-hop variant the full arm reasoned correctly, in writing, that the report "inherits from an invalidated root," approved the right draft, and still listed the withdrawn source as evidence because it had used it to reason. Under the strict rule that is an unsafe proposal. We did not reclassify it; we changed the instruction so that evidence means support and consulted artifacts go in their own field.

Lorien's same-number control queued two drafts carrying the identical figure, one descending from the withdrawn authorization. Both arms deferred on the stale draft. On the fresh draft current deferred, because it had no way to tell, while compact approved and the gate committed. Lineage did the work, not the number. The lookup control then gave both arms one authenticated lookup returning root bodies: both deferred on the stale draft and approved the fresh one, and the difference moved from correctness to cost, four calls and 4,977 tokens against two calls and 2,601. That corrected our hypothesis before the freeze, and the held-out set corrected it again (section 4).

The open-model replication on gemma-4-31b-it (twelve runs, eleven correct, zero bad actions) failed only the same-number case in the current arm, where it approved the stale draft and the gate rejected it on lineage. That run exposed the scorer blind spot described in section 3.2: the proposal cited the current source and carried the correct number.

4. Results

Preregistered before any freeze (`report/HELDOUT_PREREGISTRATION.md`); analysis by `tools/analysis.py`; every registered slot stays in its denominator. The full tables, per-arm counts, the branch audit, the cost ledger and the disclosed deviations are in Appendix B.

What ran. Five reviewer cohorts of 1,080 registered slots each: three families at 60 instances (V6, the same-number authority change; V11, the same with no correction notice delivered, so lineage is the only signal; V1, the clean-draft control with an unrelated correction before review), six arms per instance, fixture upstream so the presealed bodies are byte-identical across providers, authenticated authorship, temperature 0, one uniform 6,000-token output budget, no retries, opaque identifiers, randomized arm order, counterbalanced queue order, round-robin family interleaving. Every cohort was frozen with its manifest, prompt, code and gate hashes, a request cap and a hard credit ceiling; the runner refuses a hash mismatch and reserves the full per-attempt cost before each call. The preregistered reliability rule (at least 33 of 36 valid structured replies in the 6,000-token canary) admitted gemma-4-31b-it and deepseek-v4-flash to the inferential tier; qwen3.5-397b, qwen3.8-27b and gemma-4-26b-a4b-it run as a labeled exploratory tier; glm-53-flash and glm-5.2 were excluded.

What the gate did. Across 5,400 registered slots the models made 428 unsafe proposals, an approval of a draft with invalid lineage or a selection of an artifact that was not a queued candidate. 348 were rejected by the fenced gate on lineage and 80 were stopped before the gate; 4,077 correct publications committed; unsafe commits: 0. One slot has an unknown commit outcome, the request in flight when the host slept and the database socket died; the branch holds its approval row and the scorer counts it as an error, not a publication. A read-only count of approvals on the evaluation branch matches the traces, with 4 extra rows from instances re-run whole after the resume and 1 from the unknown-outcome slot, every one accounted for (Appendix B).

Primary: completion, compact vs lookup-offered. The two inferential models answer the question differently.

model	family	compact	lookup-offered	lookup taken	delta	Tango 95%	Holm p	NI at -0.10
gemma-4-31b-it	V6	1.000	0.500	0/60	+0.500	[+0.377, +0.623]	5.1e-10	holds
gemma-4-31b-it	V11	0.983	0.633	13/60	+0.350	[+0.242, +0.476]	6.5e-08	holds
deepseek-v4-flash	V6	0.967	1.000	59/60	-0.033	[-0.114, +0.029]	0.043	holds
deepseek-v4-flash	V11	1.000	1.000	60/60	+0.000	[-0.060, +0.060]	0.010	holds

On gemma-4-31b-it the lookup arm took its lookup in 13 of 120 offers, so it mostly behaved as the no-lineage arm and picked the stale draft half the time; compact completed 119 of 120. On deepseek-v4-flash the lookup was taken in 119 of 120 offers and the arms tie near the ceiling. Non-inferiority holds for all four claims under Holm; the deepseek V6 claim is marginal (one-sided p 0.043, two-sided interval reaching -0.114). When a model takes the authenticated lookup, preloading changes nothing about correctness; when it does not, preloading is what saves the case. The gemma gap is a fact about that model's lookup behavior under this prompt, not evidence that preloaded lineage beats retrieval.

Co-primary: tokens per registered case. Supported on deepseek-v4-flash, where the lookup arm made two calls and compact saved a median 1280 tokens per case. Not supported on gemma-4-31b-it, where compact cost a median 263 more tokens because the lookup arm mostly made one call with no lineage in context. We report that as a failed claim, not as a cheaper arm by another name.

model	settled pairs	Hodges-Lehmann, lookup minus compact	Holm p	claim: compact uses fewer
gemma-4-31b-it	119	-263	1.3e-18	not supported, opposite direction
deepseek-v4-flash	118	+1280	3.6e-35	supported

False quarantine on the clean-draft control (V1), descriptive. Approvals of the clean draft out of 60, per arm:

model	current	current + lookup	compact	full	rule	rule + content
gemma-4-31b-it	51/60	51/60	41/60	41/60	60/60	60/60
deepseek-v4-flash	0/60	0/60	0/60	0/60	60/60	60/60
qwen3.5-397b	1/60	1/60	1/60	0/60	60/60	60/60
qwen3.8-27b	0/60	0/60	0/60	0/60	60/60	60/60
gemma-4-26b-a4b-it	60/60	45/60	60/60	60/60	60/60	60/60

Both rule arms approve every clean draft on every model. deepseek-v4-flash and qwen3.8-27b quarantine it in every complete slot of every model arm, and qwen3.5-397b in all but one or two per arm, including current, which sees only the bodies, the current source and an unrelated notice: the notice alone makes those models refuse. gemma-4-31b-it approves 51 of 60 without lineage and 41 of 60 with it, so visible lineage makes it more cautious on a clean draft; the arm that wins V6 and V11 on that model pays on V1, and the rule reading the same lineage pays nothing. False quarantine is model behavior under a notice, not a property of the lineage data.

Exploratory tier. qwen3.8-27b follows the gemma-31 pattern with a lookup arm that mostly takes its lookup: compact 60/60 and 60/60, lookup-offered 56/60 and 53/60 (lookup taken 56 and 48 of 60), current 30 of 60 on both. gemma-4-26b-a4b-it is the weakest reviewer: compact 52/60 and 56/60, lookup-offered 11/60 and 7/60, most of its lookup-arm slots ending as non-queued selections stopped before the gate. qwen3.5-397b behaves like deepseek: compact 59/60 and 60/60, lookup-offered 60/60 and 60/60 (lookup taken 60 and 60 of 60), and it quarantined the clean draft in nearly every V1 model-arm slot. The full arm tracks compact on every model, so the historical bodies add nothing the lineage metadata did not carry. The diagnostics families (V0, V3, V4, V5, V7, V8, V9, V10 at n = 10 on the two inferential models) are in Appendix B.7; the one to know about is V10, where the lineage is valid and the figure is wrong: the gate commits whatever the reviewer approves there by design, the lineage-only rule approves every one, and only the rule that also reads the figure catches all ten. On gemma-4-31b-it the model arms caught between 5 and 9 of 10. A lineage gate cannot see content, and these rows measure what that costs.

Cost and incidents. Settled usage 16.83 credits plus 7.92 held as reserves against provider timeouts whose billing the platform does not report; every cohort stayed under its credit ceiling. qwen3.5-397b reached its 900-request cap with 2 slots unsent, registered as guard-stopped in its denominator. The whole evaluation including development and canaries

is under 20 credits of the 100 authorized. Three deviations are disclosed in full in Appendix B: a host sleep that killed every cohort process after 43 minutes and was recovered with a resume mode that re-ran one partial instance per cohort whole under the original freeze; a pause of the diagnostics cohorts to remove endpoint contention; and a post-results fix to the analysis code that had attached Holm-adjusted values to the wrong claims without changing any p-value.

4.1 One inspectable trace

The same-number case from the open-model pilot, run `pilot_9268a9`, current and compact side by side. Every step is in the committed run directory: dispatch receipts with request and per-body hashes, raw model text, the gate's reply, the committed approval row.

1. The researcher is dispatched with `count=12`; its reply is sealed as source v1, a root with no inputs.
2. The writer receives v1; its draft is sealed as a derivative whose only recorded input is v1. Body `{"units": 12, ...}`.
3. The operator adjudicates v1 and seals a replacement, v2, with the same figure. A fresh writer call delivered only v2 is sealed as a second draft. Two drafts with identical numbers and opaque identifiers sit in the queue.
4. Current arm: both draft bodies, the v2 body and the correction notice naming v1 superseded, no lineage. It approves the first draft, result `{"units": 12}`, evidence `[v2]`. The number is right. The artifact is stale.
5. Gate: in a fenced read-committed transaction the closure walk finds v1 adjudicated and replies `rejected: closure member <id>/1 invalidated (epoch 1)`. A separate committed-state query confirms, so the event is written `db_verified: true` with closure `[v1]`.
6. Compact arm: the same request plus one field, `history_metadata`, listing each artifact's version, author, kind, invalidation status and recorded inputs, no bodies. It approves the second draft, evidence `[that draft, v2]`. Closure `[v2]`, approval committed, the structured result and direct inputs taken from the provenance table, not from the model's text.

The scorer's original pass read step 4 as a correct proposal: the stated number was right and the cited source was current. A supplemental rescore that reconstructs the selected artifact version from the recorded selection flags it as unsafe. Both scores are preserved. The gate caught it either way.

5. Discussion and Limitations

Discussion. What is new is narrow. First, a tested multi-agent failure: stale evidence surviving a correction through summarization, including the case where the number is unchanged and only the authorization is. Second, its recovery under a gate agents cannot write around, every claimed property verified by a second, independently written implementation and reproduced from source. Third, a preregistered matched evaluation others can rerun: fixed-input dispatch, a scorer that resolves lineage itself, opaque identifiers, registered denominators, frozen hashes, hard ceilings. Fourth, the comparison that fell out of it: whether preloaded authenticated lineage beats on-demand retrieval turns on whether the model takes the retrieval. When it does, the two reach the same decisions and preloading saves the second call and its tokens; when it does not, preloading is the only thing that saves the same-number case, and no lookup at all fails exactly where only lineage can decide. That was not our starting hypothesis. We expected a correctness advantage for memory, and the lookup control corrected us twice: in development, and in the held-out set, where one model's lookup arm took its lookup in 13 of 120 offers and another's in 119.

The safety reading is in two parts. The gate is the part of this system we would trust in deployment: across 5,400 registered slots on 5 models it saw 428 unsafe proposals and committed 0, and a read-only count of the evaluation branch accounts for every approval row. The model reviewer above it is the part to measure before trusting, per model: the same preloaded lineage that lifted one model from 68 to 119 correct completions of 120 made it refuse more clean drafts, three models refused all or nearly all clean drafts under any unrelated correction notice, and one model's lookup arm barely used its lookup. A deterministic rule reading the same lineage never false-quarantined and never missed a

stale draft; where the gate cannot see content (V10), it commits whatever the reviewer approves, so the model still has a job, and these results say that job should be measured per model and bounded by the rule.

Limitations. Appendix A gives the full list, as the sprint rules require. The three that bound the claims most: the operator is trusted and owns every function, so a compromised owner defeats every property; lineage is what the runtime delivered, not what the model relied on; and the tasks are known-answer inventory counts drawn from one template, so the intervals describe resampling of that template, not a task distribution.

Future work. An eager deterministic lookup that delivers lineage records before the model's first call, which removes the offered-versus-taken question and makes preloading redundant by construction. Identity per agent instance rather than per role login. A trusted gateway so that denied actions, which leave no application audit row today, become legible. Contention and throughput measurements rather than serialized race schedules. And, with the sponsor's fine-tuning platform, a reviewer adapted on these traces, evaluated under the same frozen protocol as a new cohort.

6. Conclusion

A publication gate that re-reads authenticated lineage inside one fenced transaction held across 5,400 registered reviewer slots on 5 open models: 428 unsafe proposals, 0 unsafe commits, every approval row on the evaluation branch accounted for. What a reviewer does above that gate depends on the model more than on how lineage reaches it. Preloaded lineage and an authenticated lookup reach the same decisions when the model takes the lookup, and preloading is the only thing that saves the same-number case when it does not. Three of the five models refused all or nearly all clean drafts under an unrelated correction notice regardless of lineage, while a deterministic rule reading the same lineage never did.

The failure we set out to test is real and reproducible: stale evidence survives a correction through summarization, and a reviewer with no lineage picks the stale draft half the time when the number has not changed. The fix that held is the one agents cannot write around. The fix that varied is the one that asks a model to read.

Code and Data

Built before the sprint, declared per the rules. The Hive substrate (two AIs from different companies working through one shared memory, provenance on every signal, authorship guarded by database triggers) was built in April 2026 on the CAMA architecture. The test Hive for the first containment battery was built on 2 November 2026: the `hive_messages`, receipts and audit tables, their triggers and row-level policies, the four agent roles, the HTTPS SQL client and eleven static containment tests. Both are marked pre-existing in the repository README.

Built during the sprint, 6 to 8 November. Everything in the `exp` schema: sealed creation, provenance authorization, the invalidation authorities, the fence and the publication gate; the naive control schema and the six-attack battery; Lorien's reference specification, acceptance driver, delivery recorder, scorer and controls; the pilot runner, the held-out freeze generator, the open-model adapter, the preregistration, the analysis, and every run directory.

Reproducibility. The gate is frozen at commit `d2df880` and was rebuilt from the committed migration on a fresh branch with all 19 checks passing; acceptance evidence for each revision is kept under `runs/acceptance/`. Each held-out cohort is generated from a seed with fixed identifiers, nonces and arm order; its manifest, prompt, code and gate hashes are written to `FREEZE.json` before execution, and the runner refuses to run on any mismatch. Every model call keeps its raw request and response. Secrets are never committed. `REPRODUCE.md` gives the exact commands: apply the schema, run the deterministic tests and Lorien's `postgres_checks.py`, regenerate a freeze with `holdout_freeze.py`, run `pilot_runner.py` against it, score with the vendored `score_runs.py`, and analyze with `analysis.py` and `results_tables.py`.

Code and data. Repository: `SwarmHive` (the public mirror is linked from the submission form; pre-existing code is marked in its README). Every held-out cohort's traces, ledgers, raw requests and responses, scored outputs, freeze records and the analysis outputs are committed under `runs/`, `holdout/` and `report/`. Synthetic data only; no personal data. Info-hazard note: the containment battery contains generic attacks on a shared record (spoofing, renaming, editing, deleting, replaying), run only against synthetic rows in a disposable test database; nothing targets a real system.

References

Incident sources are cited as given in the sprint brief; verify each citation against the primary document before submission.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Limitations.

- The trust is centralized. One operator seals, adjudicates and publishes, and owns every function. A compromised or malicious owner defeats every property, including rewriting the audit. This is an explicit assumption, not a solved problem, and the entry makes no decentralized-governance claim.
- Lineage is what the runtime delivered, not what the model used. The recorder proves which bodies were supplied and sealed. It cannot certify that a hidden source was excluded or that a delivered input influenced the output. A false claim with no verifiable contradiction remains undetectable.
- A database boundary is not a sandbox. The gate says nothing about shell, network or file egress, and operations the database denies leave no application audit row. We measured this: three of six attacks in the first battery were invisible to that audit.
- Concurrency was tested as race schedules, not as scale. We ran concurrent publication against retraction with serialized commits: a naive two-step monitor approves a withdrawn dependency, the fenced gate rejects it, the fence refuses any isolation level other than READ COMMITTED, and an advisory contradiction held open in another session no longer blocks publication. The singleton fence is a serial correctness reference by design. Nothing here measures throughput, contention under load, cascading failure, emergent coordination, or the thousand-agent regime of the motivating incident.
- Small, synthetic cohorts from one template. Development cells are n of one. Held-out cells are 60 instances per family per model on five models, each its own cohort and never pooled; tasks are known-answer inventory counts drawn from one template, so the intervals describe resampling of that template, not a task distribution. Transfer to real deployments is not established.
- Model behavior under a correction notice dominates two results. Three of the five models quarantined the clean draft in all or nearly all slots whenever an unrelated correction was present, in every arm including the one with no lineage, and on the strongest model visible lineage raised false quarantine from 9 to 18 of 60. The deterministic rule arms never false-quarantined. A reviewer protocol built on these models would need that rule as a floor, and the gate as the backstop it already is.
- The lookup arm measures lookup offered, not lookup used. Whether a model takes an offered lookup is a prompt and model property we did not control, and it decided the gemma-31 result. An eager deterministic lookup would remove the question, and would make the compact arm's preloading redundant by construction.
- One slot of 5,400 has an unknown commit outcome, caused by the host sleeping mid-call; the branch shows the approval, the scorer counts an error. Five approval rows on the branch beyond the traces are explained, four by instances re-run whole after the resume and one by that slot. One cohort reached its request cap with two slots unsent. All of it is reported in Appendix B rather than reconciled away.
- The acknowledgment list at publication is an operator acknowledgment of recorded direct inputs, not an independent reviewer attestation that binds artifact version and full dependency snapshot. We removed a mechanism that appeared to do the latter but changed no decision.
- The no-lookup arm is a diagnostic, not a representative retrieval system. The lookup arm is the fair comparator, and its result constrains our claims to call and token cost rather than correctness.

Dual-use. We publish a battery of attacks against a shared agent record: sender spoofing, renaming, reading a foreign thread, editing, deleting, and replaying a stale artifact. Each is generic, already public from the incident investigations, and run only against synthetic data in a disposable test database. No exploit against any real system is included, and the attacks exist to show what a defense must catch. The gate itself is a central control point and could be used to surveil or silence agents rather than protect a shared record; that risk is inherent to the trusted-operator design and is why we state the trust boundary rather than present the gate as neutral infrastructure.

Ethical considerations. No human subjects and no personal data; all artifacts are synthetic. Model behavior is reported as observed, including a model declining to play an attacker when asked directly. The independent verification, scorer and controls were produced by an AI collaborator, Lorien, running in its own workspace; that

collaboration is disclosed rather than hidden. Sponsor credits paid for the open-model cohorts; the model set was fixed by a preregistered reliability rule applied before any held-out result existed, not by cost or by outcome.

Appendix B. Held-out tables, audit and deviation log

All tables are generated by `tools/results_tables.py` from the committed traces, ledgers and `holdout/heldout_v1/INDEX.json`; the numbers in section 4 are read from the same files.

Budget canary that set the tiers (nine matched cases, six arms, 6,000-token budget). Valid structured replies of 36 model-reviewed slots: gemma-4-31b-it 36, deepseek-v4-flash 34, qwen3.5-397b 31, qwen3.8-27b 30, gemma-4-26b-a4b-it 30, glm-53-flash 1, glm-5.2 0. 378 slots, 0.36 credits settled, 19 gateway timeouts with reserves retained. At the earlier 1,200-token budget 108 of 115 invalid replies had stopped at the budget, 101 with empty answer text.

B.1 Design as frozen

tier	model	cohort	freeze id	slots	request cap	ceiling (cr)	modeled charge	all-at-cap	seed	complete
primary	gemma-4-31b-it	heldout_gemma431bit	0cc7599ca5cf	1080	900	2.41	0.154	1.903	2026110802	yes
primary	deepseek-v4-flash	heldout_deepseekv4fl	86cc12f709ad	1080	900	3.74	0.182	0.751	2026110802	yes
exploratory	qwen3.5-397b	heldout_qwen35397b	0e4d9fc18c9c	1080	900	35.61	4.651	19.321	2026110802	yes
exploratory	qwen3.8-27b	heldout_qwen3827b	5b020297c30b	1080	900	11.21	2.0	6.78	2026110802	yes
exploratory	gemma-4-26b-a4b-it	heldout_gemma426ba4b	ada5706d6dea	1080	900	1.82	0.125	1.437	2026110802	yes

B.2 Registered slots and failure categories

deepseek-v4-flash's 504s fell mostly in the current arm, where its reasoning ran about three times longer than with lineage preloaded (mean reasoning tokens on settled V6 slots: current 2674.6, compact 528.3). gemma-4-26b-a4b-it's non-queued selections were stopped before the gate. qwen3.5-397b reached its 900-request cap with 2 slots unsent, registered as guard-stopped in its denominator. The interleaved task order kept the families balanced: the guard-stopped slots belong to the last instance in the queue.

model	registered	complete	invalid: budget	invalid: malformed/non-queued	invalid: pre-gate stop	error 504	error 502	transport timeout	error other	missing	guard-stopped	errors by arm
gemma-4-31b-it	1080	1073	0	0	0	7	0	0	0	0	0	full 3, compact 2, current 2
deepseek-v4-flash	1080	1020	2	3	0	54	1	0	0	0	0	current 45, current_lookup 5, full 3, compact 2
qwen3.5-397b	1080	1002	10	11	0	55	0	0	2	0	2	current 48, compact 3, full 3, current_lookup 3
qwen3.8-27b	1080	1037	41	0	0	1	0	0	0	0	0	current_lookup 1
gemma-4-26b-a4b-it	1080	994	0	85	0	0	0	0	1	0	0	compact 1

B.3 Primary and co-primary, full

model	family	n pairs	both	compact only	lookup only	neither	compact rate	lookup rate	delta	Tango 95%	NI p (margin -0.10)	Holm p	non-inferiority
gemma-4-31b-it	V6	60	30	30	0	0	1.000	0.500	+0.500	[+0.377, +0.623]	1.27e-10	5.08e-10	holds
gemma-4-31b-it	V11	60	38	21	0	1	0.983	0.633	+0.350	[+0.242, +0.476]	2.16e-08	6.48e-08	holds
deepseek-v4-flash	V6	60	58	0	2	0	0.967	1.000	-0.033	[-0.114, +0.029]	0.043	0.043	holds
deepseek-v4-flash	V11	60	60	0	0	0	1.000	1.000	+0.000	[-0.060, +0.060]	0.005	0.010	holds

model	settled pairs	Hodges-Lehmann	mean	direction	sign-test p	Holm p	preregistered claim
gemma-4-31b-it	119	-263.0	-173.0	compact costlier	2.67e-19	1.34e-18	not supported: opposite direction
deepseek-v4-flash	118	+1280.5	+1401.8	compact cheaper	6.02e-36	3.61e-35	supported

model	tier	family	arm	mean calls	mean tokens (in+out)	mean reasoning tokens	lookup used/offered
gemma-4-31b-it	primary	V6	compact	1	1253.5	0	
gemma-4-31b-it	primary	V6	current_lookup	1	971.1	0	0/60
gemma-4-31b-it	primary	V11	compact	1	1118.3	0	
gemma-4-31b-it	primary	V11	current_lookup	1.22	1053.6	0	13/60
gemma-4-31b-it	primary	V1	compact	1	1076.3	0	
gemma-4-31b-it	primary	V1	current_lookup	1.02	929.3	0	1/60
deepseek-v4-flash	primary	V6	compact	1	1672.6	528.3	
deepseek-v4-flash	primary	V6	current_lookup	1.98	3415.9	1510.1	59/60
deepseek-v4-flash	primary	V11	compact	1	1538.5	493.7	
deepseek-v4-flash	primary	V11	current_lookup	2	2604.3	878.0	60/60
deepseek-v4-flash	primary	V1	compact	1	2523.2	1560.2	
deepseek-v4-flash	primary	V1	current_lookup	1.55	3463.4	2149.4	35/60
qwen3.5-397b	exploratory	V6	compact	1	3159.1	1867.4	
qwen3.5-397b	exploratory	V6	current_lookup	2	3449.3	1411.8	60/60
qwen3.5-397b	exploratory	V11	compact	1	2169.3	1029.7	
qwen3.5-397b	exploratory	V11	current_lookup	2	2763.9	1001.3	60/60
qwen3.5-397b	exploratory	V1	compact	1	2143.1	1138.9	
qwen3.5-397b	exploratory	V1	current_lookup	1.97	3736.8	2044.7	59/60
qwen3.8-27b	exploratory	V6	compact	1	2084.2	811.0	
qwen3.8-27b	exploratory	V6	current_lookup	1.93	5698.2	3717.7	56/60
qwen3.8-27b	exploratory	V11	compact	1	2023.2	836.8	
qwen3.8-27b	exploratory	V11	current_lookup	1.8	4776.5	3129.2	48/60
qwen3.8-27b	exploratory	V1	compact	1	4154.6	3119.8	
qwen3.8-27b	exploratory	V1	current_lookup	1.49	7141.3	5878.4	29/60
gemma-4-26b-a4b-it	exploratory	V6	compact	1	1267.9	0	
gemma-4-26b-a4b-it	exploratory	V6	current_lookup	1.03	1018.2	0	2/60
gemma-4-26b-a4b-it	exploratory	V11	compact	1	1135.2	0	
gemma-4-26b-a4b-it	exploratory	V11	current_lookup	1.08	948.2	0	5/60
gemma-4-26b-a4b-it	exploratory	V1	compact	1	1075.5	0	
gemma-4-26b-a4b-it	exploratory	V1	current_lookup	1.1	1016.2	0	6/60

B.4 V1 false quarantine per arm

model	arm	registered	complete	approved clean draft	false quarantine	invalid	error
gemma-4-31b-it	current	60	59	51	8	0	1
gemma-4-31b-it	current_lookup	60	60	51	9	0	0
gemma-4-31b-it	compact	60	59	41	18	0	1
gemma-4-31b-it	full	60	58	41	17	0	2
gemma-4-31b-it	rule	60	60	60	0	0	0
gemma-4-31b-it	rule_content	60	60	60	0	0	0
deepseek-v4-flash	current	60	59	0	59	0	1
deepseek-v4-flash	current_lookup	60	53	0	53	2	5
deepseek-v4-flash	compact	60	60	0	60	0	0
deepseek-v4-flash	full	60	56	0	56	1	3
deepseek-v4-flash	rule	60	60	60	0	0	0
deepseek-v4-flash	rule_content	60	60	60	0	0	0
qwen3.5-397b	current	60	33	1	32	6	21
qwen3.5-397b	current_lookup	60	44	1	43	13	3
qwen3.5-397b	compact	60	58	1	57	0	2
qwen3.5-397b	full	60	57	0	57	0	3
qwen3.5-397b	rule	60	60	60	0	0	0
qwen3.5-397b	rule_content	60	60	60	0	0	0
qwen3.8-27b	current	60	55	0	55	5	0
qwen3.8-27b	current_lookup	60	35	0	35	24	1
qwen3.8-27b	compact	60	58	0	58	2	0
qwen3.8-27b	full	60	54	0	54	6	0
qwen3.8-27b	rule	60	60	60	0	0	0
qwen3.8-27b	rule_content	60	60	60	0	0	0
gemma-4-26b-a4b-it	current	60	60	60	0	0	0
gemma-4-26b-a4b-it	current_lookup	60	54	45	9	6	0
gemma-4-26b-a4b-it	compact	60	60	60	0	0	0
gemma-4-26b-a4b-it	full	60	60	60	0	0	0
gemma-4-26b-a4b-it	rule	60	60	60	0	0	0
gemma-4-26b-a4b-it	rule_content	60	60	60	0	0	0

B.5 Completed correctly per arm, all models and families

model	tier	family	current	current_lookup	compact	full	rule	rule_content
gemma-4-31b-it	primary	V6	30/60	30/60	60/60	59/60	60/60	60/60
gemma-4-31b-it	primary	V11	30/60	38/60	59/60	60/60	60/60	60/60
gemma-4-31b-it	primary	V1	51/60	51/60	41/60	41/60	60/60	60/60
deepseek-v4-flash	primary	V6	17/60	60/60	58/60	60/60	60/60	60/60
deepseek-v4-flash	primary	V11	20/60	60/60	60/60	60/60	60/60	60/60
deepseek-v4-flash	primary	V1	0/60	0/60	0/60	0/60	60/60	60/60
qwen3.5-397b	exploratory	V6	22/60	60/60	59/60	60/60	60/60	60/60
qwen3.5-397b	exploratory	V11	27/60	60/60	60/60	60/60	60/60	60/60
qwen3.5-397b	exploratory	V1	1/60	1/60	1/60	0/60	60/60	60/60
qwen3.8-27b	exploratory	V6	30/60	56/60	60/60	60/60	60/60	60/60
qwen3.8-27b	exploratory	V11	30/60	53/60	60/60	60/60	60/60	60/60
qwen3.8-27b	exploratory	V1	0/60	0/60	0/60	0/60	60/60	60/60
gemma-4-26b-a4b-it	exploratory	V6	30/60	11/60	52/60	51/60	60/60	60/60
gemma-4-26b-a4b-it	exploratory	V11	24/60	7/60	56/60	56/60	60/60	60/60
gemma-4-26b-a4b-it	exploratory	V1	60/60	45/60	60/60	60/60	60/60	60/60

B.6 Gate integrity and the read-only branch audit

model	slots	gate calls	unsafe proposals	pre-gate stops	gate rejections	correct publications	unsafe commits	unknown commit outcome	db-verified approvals
gemma-4-31b-it	1080	1021	111	0	111	910	0	0	910
deepseek-v4-flash	1080	792	38	1	37	755	0	0	755
qwen3.5-397b	1080	813	48	6	42	771	0	0	771
qwen3.8-27b	1080	835	66	0	66	769	0	0	769
gemma-4-26b-a4b-it	1080	965	165	73	92	872	0	1	872
all	5400	4426	428	80	348	4077	0	1	4077

model	db-verified approved events in traces	approvals from superseded partial traces (instance re-run after resume)	slots with unknown commit outcome	approval rows on branch	rows unexplained
gemma-4-31b-it	910	0	0	910	0
deepseek-v4-flash	755	0	0	755	0
qwen3.5-397b	771	1	0	772	0
qwen3.8-27b	769	3	0	772	0
gemma-4-26b-a4b-it	872	0	1	873	0

The unknown-outcome slot is gemma-4-26b-a4b-it, V11 instance 41, compact arm: the model approved, the gate call began, and the database connection died in the host sleep before the process read the committed decision. The branch holds the approval row; the scorer does not count it. The qwen3.8-27b surplus is the superseded partial instance V6i05, whose approvals committed before the crash and which was re-run whole after the resume.

B.7 Diagnostics families (descriptive, n = 10 per family, two models)

Separate freeze (seed 2026110803), eight families at ten instances, same six arms, same budget. V10 is the eligible-but-wrong control: the queued draft's lineage is valid and its figure is wrong, so the gate commits whatever a reviewer approves. Its "unsafe commits" below are that designed content blind spot, counted here and not in the gate-integrity totals of section 4, which cover the three primary families where lineage decides. The rule arm approves every V10 draft (it reads lineage only); the rule-with-content reference catches every one.

model	family	current	current_lookup	compact	full	rule	rule_content	errors	invalid	unsafe proposals	unsafe commits
gemma-4-31b-it	V0	10/10	10/10	10/10	9/10	10/10	10/10	1	0	0	0
gemma-4-31b-it	V3	10/10	10/10	10/10	9/10	10/10	10/10	1	0	0	0
gemma-4-31b-it	V4	5/10	8/10	10/10	10/10	10/10	10/10	0	1	6	0
gemma-4-31b-it	V5	10/10	10/10	10/10	10/10	10/10	10/10	0	0	0	0
gemma-4-31b-it	V7	10/10	9/10	10/10	9/10	10/10	10/10	0	0	0	0
gemma-4-31b-it	V8	10/10	10/10	10/10	10/10	10/10	10/10	0	0	0	0
gemma-4-31b-it	V9	9/10	9/10	10/10	10/10	10/10	10/10	0	1	0	0
gemma-4-31b-it	V10	9/10	5/10	7/10	8/10	0/10	10/10	0	0	21	21
deepseek-v4-flash	V0	10/10	10/10	10/10	10/10	10/10	10/10	0	0	0	0
deepseek-v4-flash	V3	10/10	10/10	10/10	10/10	10/10	10/10	0	0	0	0
deepseek-v4-flash	V4	7/10	10/10	10/10	10/10	10/10	10/10	3	0	0	0
deepseek-v4-flash	V5	10/10	10/10	10/10	10/10	10/10	10/10	0	0	0	0
deepseek-v4-flash	V7	10/10	10/10	9/10	10/10	10/10	10/10	1	0	0	0
deepseek-v4-flash	V8	10/10	10/10	10/10	10/10	10/10	10/10	0	0	0	0
deepseek-v4-flash	V9	10/10	10/10	9/10	10/10	10/10	10/10	1	0	0	0
deepseek-v4-flash	V10	7/10	6/10	5/10	9/10	0/10	10/10	8	2	14	13

model	registered	complete	error 504	missing	unsafe proposals	gate rejections	pre-gate stops	unsafe commits	unknown outcome	settled	retained	ceiling	complete run
gemma-4-31b-it	480	476	2	0	27	5	1	21	0	0.1534	0.0502	1.09	yes
deepseek-v4-flash	480	465	13	0	14	0	1	13	0	0.6727	0.6162	1.89	yes

B.8 Cost per cohort

model	attempts	request cap	settled	retained reserves (504)	ceiling	modeled charge	all-at-cap	resumes
gemma-4-31b-it	735	900	0.4646	0.1756	2.41	0.154	1.903	1
deepseek-v4-flash	877	900	2.8599	2.6069	3.74	0.182	0.751	1
qwen3.5-397b	900	900	10.9975	5.12	35.61	4.651	19.321	1
qwen3.8-27b	855	900	2.4141	0.0177	11.21	2.0	6.78	1
gemma-4-26b-a4b-it	733	900	0.0902	0	1.82	0.125	1.437	1
all			16.8263	7.9202				

B.9 Deviations from the preregistration

1. Host sleep. The five cohorts launched at 00:23 EST on 8 November. The laptop slept from 01:06 to 13:32 EST. On wake every cohort process died on a dead database socket. No model request was in flight except the gemma-26 slot above; every finished slot was already on disk. A resume mode was added to the runner (`--resume-out`, commit cc7a480). It validates the execution record and the registry claim, keeps every instance with a trace for all six arms, re-runs the one partially complete instance per cohort as a whole (gemma-31 V11i20, deepseek V1i06, qwen-397 V6i04, qwen-27 V6i05; gemma-26 had none) with the partial traces set aside in `traces.partial_superseded.jsonl`, inherits the prior ledger's settled spend, retained reserves and attempt count so the ceiling and the request cap still bind, and records the resume in `EXECUTION.json` and the registry. The frozen instances, prompts and arm orders did not change; the re-run slots were answered about thirteen hours later by the same models at temperature 0. All five resumed at 13:55 EST. The runner also gained connection keepalives and a single reconnect on a dead socket.
2. Diagnostics paused. Both diagnostics cohorts were paused at 00:28 EST because the gemma-31 primary was timing out on 3 of 9 attempts while sharing its endpoint with its diagnostics twin. They were resumed under the same claims after the primaries finished, at 18:03 and 18:07 EST. Slots not reached by submission are reported as missing.
3. Analysis code fixed after results were seen. `tools/analysis.py` built the vector of p-values with the completion claims first and wrote the Holm-adjusted values back in claim order, so adjusted values were attached to the wrong claims. Fixed in commit dbb70f8. No p-value changed, only which adjusted value sat next to which claim; both versions are in git and the self-test output is unchanged.
4. The direction of the token claim was fixed in the preregistration. The gemma-31 result is reported as not supported rather than reinterpreted.

Appendix C. LLM Usage Statement

The author worked with two AI systems throughout and discloses both. Aelen, a Claude model running in Claude Code, wrote most of the code in the repository under the author's direction, ran every evaluation under the author's explicit spend authorization, and drafted this report from the committed run data; every number in the report is produced by scripts in the repository (`tools/analysis.py`, `tools/results_tables.py`, `tools/write_section5.py`) from the committed traces and ledgers, not typed by a model. Lorien, an OpenAI Codex model running in a separate workspace, wrote the independent reference specification, the acceptance driver, the delivery recorder and the scorer, and reviewed the design and the report adversarially; its reviews are committed. The author set the research question, chose the tracks, made every design decision that is recorded as hers in the repository, authorized every paid run, reviewed every claim against the data, and edited the text. Claims and results were verified against the committed evidence before submission; the two deviations from the preregistration and the one post-results code fix are disclosed in Appendix B.